

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Science — Semester I Examination, 2023-24

Paper No. MTB-101: Calculus-I

Time: 3 Hours

Maximum Marks: 70

Ref. No.: 067931

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1. (Compulsory — 2 marks each.)

- (a) Find the least $m \in \mathbb{N}$ for the sequence $x_n = \frac{3n+1}{7n-4}$ such that $\left| \frac{3n+1}{7n-4} - \frac{3}{7} \right| < \varepsilon$ for all $n > m$.
- (b) Check whether $\{x_n\}$ defined by $x_1 = \sqrt{7}$, $x_{n+1} = \sqrt{7+x_n}$ is monotonically increasing or decreasing.
- (c) Examine convergence of $\sum_{n=1}^{\infty} \frac{n}{n(n+1)(n+3)}$.
- (d) If $\sum u_n$ converges with positive terms, prove $\sum u_{2n}$ converges.
- (e) Let $f : I \rightarrow \mathbb{R}$, $\lim_{x \rightarrow c} f(x) = \ell$. Show $\lim_{x \rightarrow c} |f(x)| = |\ell|$.
- (f) Give an example of f, g not continuous at c but fg continuous at c .
- (g) $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(x+y) = f(x)f(y)$. If f is differentiable at 0, prove $f'(c) = f(c)f'(0)$.
- (h) Let $f : I \rightarrow \mathbb{R}$ be differentiable at c with $f'(c) < 0$. Show f is decreasing at c .
- (i) If $c_0 + \frac{c_1}{2} + \frac{c_2}{3} + \cdots + \frac{c_n}{n+1} = 0$, show $c_0 + c_1x + \cdots + c_nx^n = 0$ has a root in $(0, 1)$.
- (j) Let $y = e^x(\cos x + \sin x)$. Prove $y'' + 4y = 0$.
- (k) Draw the graph of $r = 2 \sin(2\theta)$.

Question 2.

- (a) Let $\{x_n\}, \{y_n\}$ converge to x, y (non-zero). Show $\lim_{n \rightarrow \infty} \frac{x_n}{y_n} = \frac{x}{y}$ provided $y \neq 0$. [6]
- (b) Let $x_{n+1} = \frac{1}{2}(x_n + y_n)$, $y_{n+1} = \sqrt{x_n y_n}$ for $n \geq 1$, $x_1, y_1 > 0$. Prove $\lim x_n = \lim y_n$. [6]

Question 3.

- (a) Let $\sum u_n, \sum v_n$ be positive-term series with $\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \ell \neq 0$. Prove they converge or diverge together. [6]
- (b) Test convergence of $\sum_{n=0}^{\infty} \frac{n^2 - 1}{n^2 \cdot 2^n}$. [3]
- (c) Give ε - δ definition of $\lim_{x \rightarrow c} f(x) = \ell$. Give geometrical representation. Show $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$. [3]
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Question 4.

- (a) Let $f : I \rightarrow J$ continuous at c , $g : J \rightarrow \mathbb{R}$ continuous at $f(c)$. Show $g \circ f$ is continuous at c . [6]
- (b) Explain differentiability of f at c . If f differentiable at c , show f continuous at c . Give an example of a function continuous but not differentiable at a point in $[-4, 4]$. [6]
- (c) Let $f : [a, b] \rightarrow \mathbb{R}$ be differentiable with $f'(a) \neq f'(b)$. If u lies between $f'(a)$ and $f'(b)$, prove $\exists c \in (a, b)$ with $f'(c) = u$. [6]
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Question 5.

- (a) State Rolle's theorem. Verify it for $f(x) = x \sin \frac{1}{x}$ ($x \neq 0$), $f(0) = 0$. [6]
- (b) State and prove Lagrange's Mean Value Theorem. Verify with an example. [6]
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Question 6.

- (a) Find the Maclaurin expansion of $y = e^x \sin x$ in powers of x . [6]
- (b) Find the n -th derivative of $f(x) = \log(5x + 3) \cdot \cos(3x + 5)$. [6]
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Question 7.

- (a) Find the area under $y = x^2$ from $x = \frac{1}{2}$ to $x = 2$ using limit of a sum. [6]
- (b) Find asymptotes of $y = \frac{x^2 + 2x - 1}{x}$ and trace the curve. [6]
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